

Thomas Gao

t25gao@uwaterloo.ca · linkedin.com/in/thomas-gao-1a2b3c · github.com/Thomasg101

SUMMARY

Civil Engineering student with field and analytical experience spanning **building science, structural restoration, deep foundations, and pavement materials**. Has modelled envelope thermal performance in **Flixo, THERM, HEAT3, and Ubakus**, coordinated **deep foundation and support-of-excavation** work at Pape Station on the Ontario Line, and independently run **MTO highway** quality control. Manages work from **tender review through substantial performance**, and builds software that makes engineering teams faster.

EXPERIENCE

Building Envelope Specialist (Co-op)

Entuitive

May 2026 – August 2026, Vancouver, BC

- Performed **2D and 3D thermal modelling** in **Flixo, THERM, HEAT3, and Ubakus** to quantify thermal bridging at slab edges, parapets, window transitions, balconies, and cladding clips — producing effective U-values and **linear (ψ) and point (χ) transmittance** for **BC Energy Step Code, NECB, and ASHRAE 90.1** compliance.
- Authored **Site Review Reports** documenting construction conformance to drawings and specifications, with annotated photo logs, deficiency registers, and follow-up tracking.
- Completed **Building Condition Assessments** and **Depreciation Reports** for **BC Housing** and private clients — component inventories, condition ratings, remaining service life, and 30-year capital renewal cost models.

Engineering Technician (Co-op)

Brown Beattie Building Science Engineering

September 2025 – December 2025, Richmond Hill, ON

- Conducted site inspections for **building restoration, waterproofing, and concrete repair**, verifying surface preparation, **CSP profiles**, adhesion, and membrane thickness per **OBC, CSA**, and manufacturer standards.
- Prepared comprehensive **Site Visit Reports** with technical observations, field test data, and deficiency logs supporting engineering review and contract compliance.
- Oversaw multiple envelope rehabilitation projects from **tender review through substantial performance**, coordinating mock-ups, submittal reviews, and shop drawing review across envelope and drainage systems.

Project Coordinator (Co-op)

Green Infrastructure Partners

January 2025 – May 2025, Toronto, ON

- Coordinated **deep foundation and shoring** operations for **Pape Station (Ontario Line)** — diaphragm wall construction, caisson drilling, soldier pile and lagging, **O-cell load testing**, and tieback anchor testing.
- Interpreted **geotechnical and structural design drawings**, sequencing excavation and **support of excavation (SOE)** activities to **MTO, CSA**, and **Ministry of Labour** requirements, and prepared field reports, as-builts, and quantity summaries.

Quality Control Technician

GIP Construction Engineering

June 2024 – September 2025, Thunder Bay, ON

- Performed field and laboratory QC testing on **Superpave** and **CIREAM** materials to **MTO, OPSS 1150, ASTM**, and **AASHTO** standards, including Lottman (AASHTO T283), Marshall stability and flow, and bulk specific gravity.
- Independently managed QC operations** for **MTO Highway 17 and Highway 61** rehabilitation, and supported mix optimization through gradation, binder content, and foaming parameter analysis.

TECHNICAL PROJECTS

- ProjectSearch** (Python, FastAPI, SQLite, three.js; built at Entuitive) — internal web app making the firm's photo and document archive searchable in plain English: hybrid semantic + BM25 retrieval with AI reranking, vision-model tagging of components, materials, and deficiencies, and a GPS-driven **3D building view**. Cut report-production retrieval from hours to minutes; **under review for firm-wide implementation**.
- 2D Frame & Truss Analysis Tool** (Python, NumPy, Matplotlib) — **direct stiffness method** solver returning displacements, member end forces, and reactions with axial, shear, and moment diagrams; validated against hand calculations and SAP2000.

EDUCATION

Bachelor of Applied Science, Civil Engineering

University of Waterloo

Expected 2028, Waterloo, ON

- Cumulative GPA 4.0 · President's Scholarship of Distinction, 2024

SKILLS

Software: AutoCAD, Revit, Civil 3D, Bluebeam Revu, SAP2000, S-FRAME, Excel (advanced), PVSyst

Programming: Python, MATLAB, SQL, VBA

Building Science: Flixo, THERM, HEAT3, Ubakus, WUFI; thermal bridging and condensation analysis, air and vapour barrier systems, EIFS, PMMA and SBS waterproofing

Codes & Standards: NBC, OBC, BCBC, BC Energy Step Code, NECB, ASHRAE 90.1, CSA A23.1 / A23.3 / S16 / A440, MTO OPSS, ASTM, AASHTO